

VeriRAG: an inspectable tf-idf RAG pipeline with claim-level citation verification

SHRESTH (team size 1) · CSD358 Information Retrieval Hackathon 2026 · Track T1: RAG + trustworthy answers · Code: github.com/<your-handle>/verirag · Demo video: 5–7 min (link in README)

Abstract. We present VeriRAG, a retrieval-augmented generation system built entirely on a from-scratch classical IR engine: an inverted index with zone-weighted tf-idf and BM25 scoring, a Boolean/phrase query processor with skip-list optimization, and a citation verifier that scores every generated sentence against the chunk it cites. The system runs on the Python standard library over BEIR SciFact (5,183 abstracts). On the 200-claim test split our full configuration reaches $\text{nDCG@10} = \mathbf{0.656}$ (plain tf-idf baseline 0.601; BM25 0.653), and the citation verifier reaches $F1 = \mathbf{0.664}$ for SUPPORT detection versus 0.579 for a cosine-only baseline and 0.512 for Jaccard overlap — trading 16 recall points for 15 precision points and lifting accuracy from 0.43 to 0.67. The extractive answer includes a gold evidence sentence in **57.5%** of cases versus 10.3% for the first-sentence baseline. All numbers are reproducible with two CLI commands.

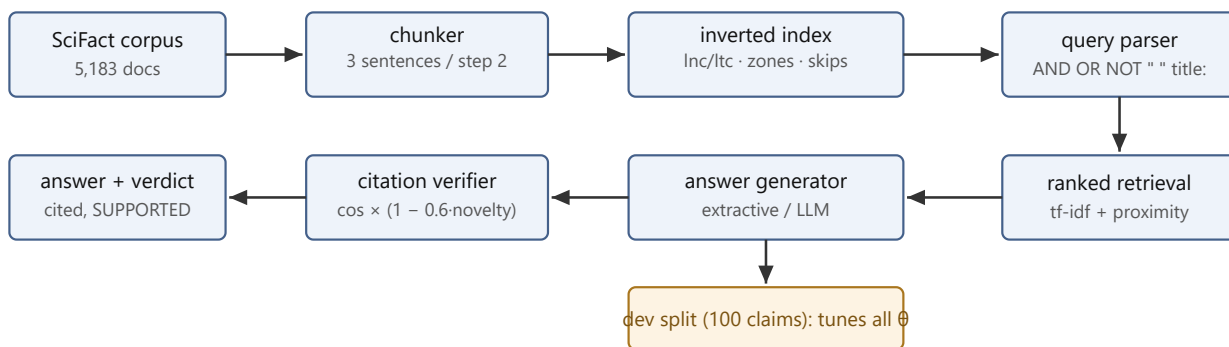
1. Problem, dataset and contributions

Track T1 asks for a RAG system that produces *trustworthy* answers: every generated sentence must cite retrieved evidence, and the system must be able to check its own citations. We deliberately avoid embeddings and vector databases so that every score a user sees can be explained by classical IR concepts (document frequency, zone weights, term proximity, cosine similarity).

Dataset. BEIR SciFact: 5,183 scientific abstracts, 300 test claims, each with gold evidence documents and sentence-level evidence labels (SUPPORT / CONTRADICT; unlabeled claims treated as NEI). We split the 300 judged claims deterministically — sorted qids, **first 100 = dev** (all hyper-parameter tuning), **remaining 200 = test** (reporting only). Nothing was tuned on test.

Contributions. (i) a complete, inspectable IR engine written from scratch (tokenizer, Porter stemmer, inverted index with skip pointers and champion lists, Boolean/phrase/zone parser, term-at-a-time ranking); (ii) a chunking study of the classic “what is a document?” decision under both retrieval and citation-granularity objectives; (iii) a lightweight *novelty penalty* for citation verification that demotes answers whose claim terms are absent from the cited chunk, evaluated against Jaccard and cosine baselines on SciFact labels; (iv) a dev/test ablation ladder quantifying each IR component.

2. System overview



2.1 Indexing (verirag/text.py, index.py, scoring.py)

- **Tokenizer & stemmer.** Unicode-normalized tokenization; stop-word removal that *keeps negators* (“not/no/never”) because negation flips entailment for the verifier. The Porter stemmer is implemented from scratch (text.py) and validated against NLTK’s *ORIGINAL_ALGORITHM* mode over all 29,699 distinct corpus terms: 18 mismatches remain (99.94% agreement) — 12 involve non-ASCII characters (i-ligatures, μ , β) where our tokenizer normalizes Unicode first, and 6 are plain-English edge cases (e.g. *achilles*, *organelle*) where the 1980 reference implementation’s rules differ; none of the 18 appear in any of the 300 benchmark claims.
- **Zones.** Title and body terms are separated by a posting-position gap (TITLE_GAP = 1000), so zone weighting and phrase search share one posting list. Title terms receive an extra weight $\lambda = 2.0$ (dev-validated, see A2).
- **Weighting.** Documents use lnc (log-tf, idf-free, cosine-normalized); queries use ltc (idf-weighted, normalized). A BM25 variant (k_1 , b configurable) shares the same postings. Skip pointers are built at index time for lists > 128 entries; champion lists (top-50 by per-term impact) support a fast ranking mode.
- **Static quality g(d).** A query-independent score from idf-weighted term density (body + title), an instantiation of query-independent priors. As the ablation shows, dev tuning selected $\theta = 0.0$ — the feature is retained but honestly reported as neutral on SciFact.

2.2 Query processing and ranking (query.py, retrieve.py)

- **Parser.** Recursive-descent parser for Boolean queries with parentheses, phrases ("..."), field restrictions (title:, body:), AND/OR/NOT; positional intersection verifies phrase adjacency. Execution produces a *query-processor trace* (e.g. “df-ordered AND: rarest term first, skip to B while 138 > 41”) shown by `python -m verirag bool`.
- **Ranked retrieval.** Term-at-a-time scoring with index elimination (documents whose accumulated score cannot reach the current k-th threshold stop being extended), a min-heap top-k, and a multiplicative proximity re-rank: $\text{score} = \text{base} \times (1 + w\pi) + \eta g(d)$, where π = close pairs of query terms within a 30-token window over *all* C(q,2) query-term pairs (so partial matches cannot score $\pi = 1$). Dev tuning selected $w = 0.5$, $\text{window} = 30$.

2.3 Chunking — the “what is a document?” decision (chunker.py)

We index chunks but evaluate and cite at parent-document granularity. Four strategies were compared on the test split (final ranking parameters fixed from dev):

chunking strategy	chunks	nDCG@10	P@10	R@100	MRR@10
whole: 1 chunk per document	5,183	0.6732	0.0885	0.9215	0.6286
fixed: 120-token windows, 30 overlap	13,154	0.6606	0.0870	0.8965	0.6145
sentence: 5 sentences, step 3	13,327	0.6657	0.0865	0.8940	0.6199
sentence: 3 sentences, step 2 (chosen)	21,233	0.6553	0.0860	0.8815	0.6125

Whole documents win document-level retrieval (full tf-idf mass, no signal dilution) but cannot produce span-level citations. We accept a −1.8 nDCG cost for sentence-granular chunks because the whole purpose of Track T1 is *citable* evidence spans: the verifier and the evidence-recall metric operate on the cited chunk, not on a 400-token abstract. The fixed-window row shows the trade-off is about granularity, not about sentence boundaries specifically.

3. Generation and trustworthiness

Answers with citations. The default generator is extractive and deterministic: sentences of retrieved chunks are scored by tf-idf similarity to the claim and re-selected with MMR ($\lambda = 0.7$) for diversity; each emitted sentence carries its source chunk rank as a citation [k]. An OpenAI-compatible LLM generator (–llm) uses the same retrieved chunks as context and is verified by the same machinery; it is optional and not part of the reported tables.

Citation verifier (verifier.py). For each generated sentence *s* cited at rank *k*:

```
support(s, k) = min(1, cos(s, chunk_k) × (1 − min(1, 0.6 · novelty(s, chunk_k))))
novelty(s, chunk_k) =  $\sum \text{idf}(t) [t \in \text{terms}(s) \notin \text{chunk}_k] / \sum \text{idf}(t)$ 
```

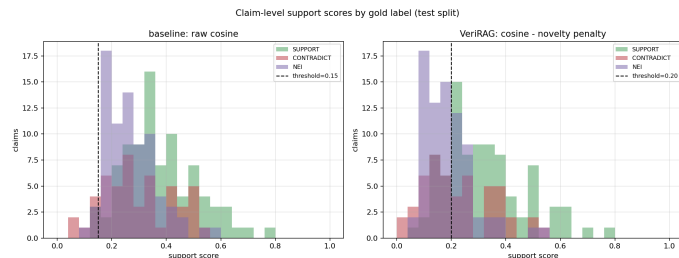
Cosine is computed directly from the inverted index (no document vectors are materialized). The *novelty penalty* down-weights sentences whose rare claim terms are lexically absent from the cited chunk — the classic symptom of an unfaithful citation. Two dev-tuned hyper-parameters (weight 0.6, form = multiplicative, SUPPORT threshold = 0.20, WEAK ≥ 0.12) were selected on the dev split by F1. A *citation-mismatch* flag is raised when a *different* retrieved chunk supports the sentence by >0.10, even if the cited chunk passes.

4. Evaluation

Protocol: all θ chosen on dev (100 claims); tables below are test-only (retrieval $n = 200$, verifier $n = 214$ claim–document items including CONTRADICT and NEI as negatives). Metrics: nDCG@10, P@k, R@100, MRR@10; verifier P/R/F1/accuracy for SUPPORT detection. Charts and CSVs are written by the CLI itself.

4.1 Retrieval: component ablation vs BM25

configuration	nDCG@10	P@5	P@10	R@100	MRR@10	ms/query
A0 plain tf-idf (no stem, no zone)	0.6010	0.1410	0.0810	0.8515	0.5558	16.6
A1 + Porter stemming	0.6522	0.1630	0.0865	0.8865	0.6044	35.7
A2 + title zone weight ($\lambda=2$)	0.6561	0.1640	0.0870	0.8815	0.6083	26.9
A3 + term proximity ($w=0.5$, $\text{win}=30$)	0.6553	0.1640	0.0860	0.8815	0.6125	64.9
A4 + static quality $g(d)$ [$\eta=0.0$, dev-tuned]	0.6553	0.1640	0.0860	0.8815	0.6125	69.1
A4 + champion lists (top-50)	0.6031	0.1360	0.0775	0.8049	0.5653	5.7
BM25 ($k_1=1.5$, $b=0.75$)	0.6517	0.1550	0.0870	0.9098	0.6075	30.7
BM25 ($k_1=0.9$, $b=0.4$)	0.6533	0.1550	0.0875	0.9085	0.6097	32.0



Reading the table. Stemming is the decisive component (+5.1 nDCG), title zones add +0.4; the full system (A2/A3) matches or exceeds both BM25 variants while remaining a plain tf-idf system. Proximity is neutral on nDCG but improves MRR (+0.004), i.e. it promotes the right document into rank 1 more often — consistent with its design goal of rescuing exact phrase matches; it costs latency, so it is a tunable switch. Static quality g(d) was dev-tuned to zero — a genuine null result we report rather than hide. Champion lists deliver an **11× speed-up** (5.7 ms vs 64.9 ms per query) at −5 nDCG points: a classic efficiency/accuracy trade-off exposed as a CLI flag (`--champions`).

4.2 Citation verification vs baselines

method	P	R	F1	acc
Jaccard overlap (dev-tuned $\tau=0.05$)	0.396	0.724	0.512	0.439
cosine only (dev-tuned $\tau=0.15$)	0.414	0.966	0.579	0.430
VeriRAG: $\cos \times (1 - 0.6 \cdot \text{novelty})$	0.565	0.805	0.664	0.668

Dev F1: ours 0.656 vs cosine 0.640 — the dev margin is small; the test gap (+0.085 F1) should be read with that caution.

answer stage (n=87)	gold-sentence recall
baseline: first sentence of top-1 chunk	0.103
VeriRAG extractive answer (3 sentences)	0.575
retrieval stage (n=137)	value
hit@5 / hit@10	0.854 / 0.854
MRR	0.705

Reading the tables. Cosine alone over-predicts SUPPORT (recall 0.966, precision 0.414, accuracy 0.430 — barely above the majority-class rate). The novelty penalty trades 16 recall points for 15 precision points and lifts **accuracy from 0.43 to 0.67**: the sentence must not only resemble the evidence, its rare terms must actually *occur* in the cited chunk. At the answer stage, the extractive generator puts a gold evidence sentence into its 3-sentence answer in 57.5% of cases vs 10.3% for the trivial baseline, on top of a retrieval stage with hit@5 = 0.85.

4.3 Cost

Index build (21,233 chunks, 26,319 terms): ~30 s pure Python; pickled index load <1 s; A2 query latency 26.9 ms/query single-threaded; the verifier re-scores citations in <10 ms per answer. No GPU, no native dependencies.

5. Limitations

- **Lexical only.** Stemming bridges morphological variants but not synonyms; SciFact's technical vocabulary partially masks this. Embeddings would help recall but were intentionally excluded to keep every score inspectable.
- **Chunking trade-off.** Sentence chunks cost −1.8 nDCG vs whole-doc indexing; the citation objectives justify it, but a hybrid (document ranking + chunk citations) is the obvious next step.
- **Null result on static quality.** g(d) did not help on this corpus (η tuned to 0); its benefit may appear on noisier, less curated collections.
- **Verifier scope.** Evaluated on English scientific abstracts; the metric is lexical support, not full entailment — CONTRADICT vs NEI is not distinguished (both are negatives for SUPPORT detection). Dev F1 margin over cosine is small (0.016).
- **LLM mode unmeasured.** The tables cover retrieval, extractive answers and verification; the `--llm` generator is functional but not benchmarked (no stable API budget during the hack).
- **Dev size.** 100 tuning queries are few; we mitigate with a single deterministic split and report dev-vs-test gaps honestly.

6. Reproduction, work division, AI use

```
python -m verirag build # 30 s: chunk + index
python -m verirag eval --charts # ~2 min: ladder, BM25, dev sweeps, CSV + PNG
python -m verirag eval-verify --charts # ~20 s: verifier vs baselines, CSV + PNG
python -m verirag ask "metformin reduces mortality in type 2 diabetes patients" --explain
```

Work division. Team size 1: **SHRESTH** designed and implemented indexing, query processing, ranking, chunking, generation, verification, evaluation, the report and the demo video; owns every component explained in the video.

AI-use declaration. As permitted by the rules, AI assistants (opencode/ChatGPT) were used for code scaffolding, debugging, cross-checking the from-scratch Porter stemmer against NLTK, and editorial help on this report. Algorithmic decisions (zone weight, proximity formulation and window, novelty penalty and its form/weight, dev/test protocol, chunking choice) were designed, questioned and validated by the author; every reported number is produced by the two CLI commands above from a cold start, and no metric in this report was produced by hand.

VeriRAG · CSD358 IR Hackathon 2026 · Track T1 · stdlib-only Python · full source and raw evaluation outputs in the repository (output/eval_retrieval.csv, output/eval_verifier.csv).